Western New England University

College of Engineering

Electrical and Computer Engineering Department

Lab 05: Frequency Response of the Common Source Amplifier with Feedback

Question	Score
Lab Notebook	/ 100
Total	/ 100

1 Assignment Goals

In this lab, our goal is to modify the common source amplifier used in Lab 04 and apply feedback to the circuit. Specifically, negative feedback is created by adding a resistor from the drain of Q_1 to the gate of Q_1. Feedback achieves many objectives including stabilizing the DC bias point which no longer requires a dedicated bias voltage. Students will observe that the bandwidth will increase, but at the cost of overall circuit gain. However, students will also observe that feedback stabilizes the gain as it only dependent on feedback resistor and in this case, the signal generator resistor. will be implemented using the ALD1105 MOSFET to analyze the frequency response. LTSpice will be used to validate measurement techniques.

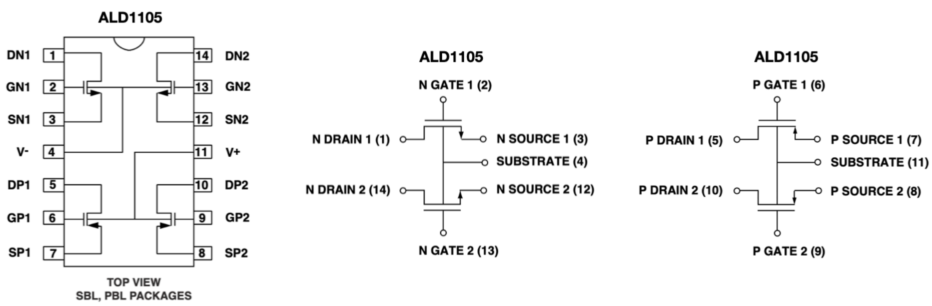
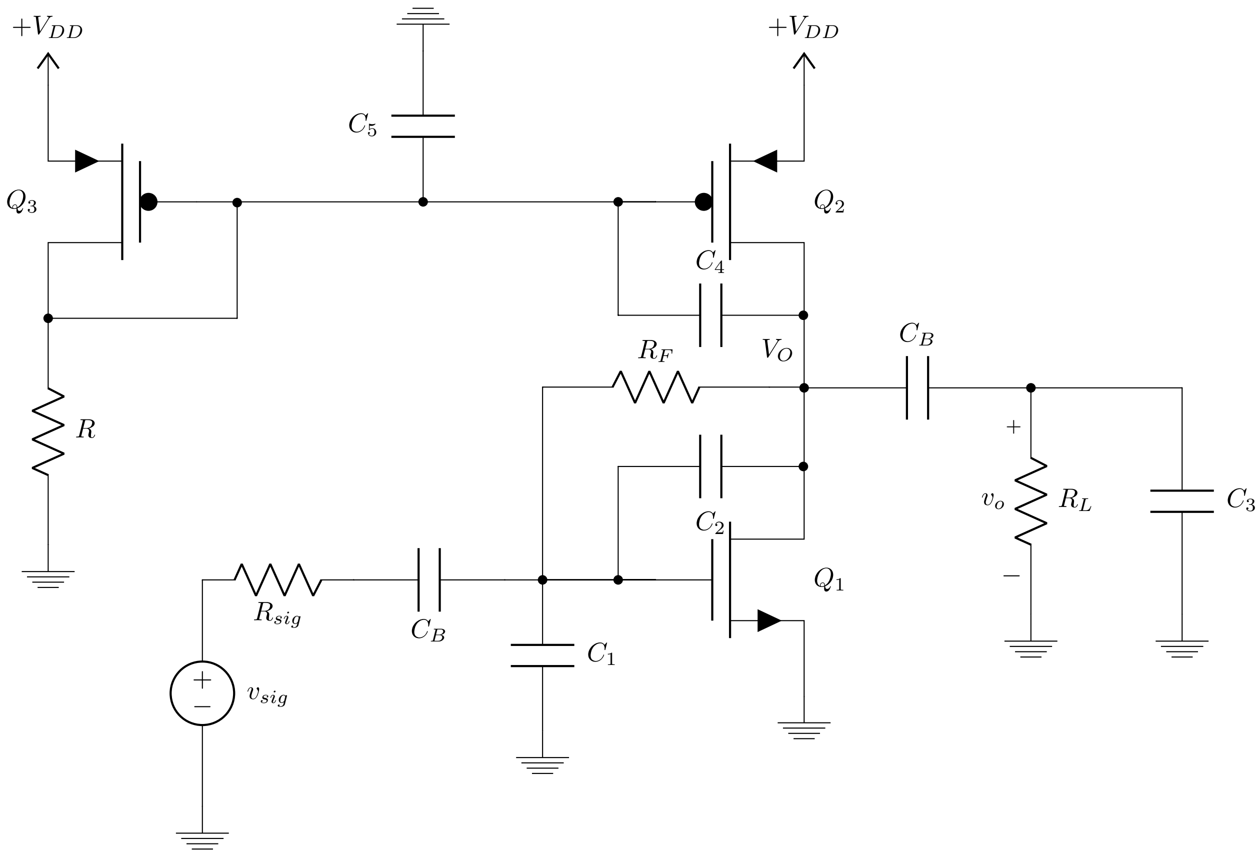


Figure 1: The pinout and block diagram of the ALD1105, **Note: V^- is the body of the NMOS devices which is connected to lowest potential and V^+ is the body of the PMOS devices which is connected to the highest potential.**

2 Experiment 1: DC Operating Point

Using the circuit shown below with the parameter values given, adjust the potentiometer such that the total current is equal to $I_{Total} = 1\text{mA}$.

- Find the dc operating point by measuring the node voltages and calculating the currents. Complete the summary table.



$V_{DD} = 10\text{V}$ $\quad C_B = 0.022\mu\text{F}$ $\quad R = 25\text{k}\Omega$ Potentiometer

$R_F = 330\text{k}\Omega$ $\quad R_{sig} = 100\text{k}\Omega$ $\quad R_L = 10\text{M}\Omega$ (Probe)

$C_1 = 10\text{pF}$ $\quad C_2 = 3.3\text{pF}$ $\quad C_3 = 22\text{pF}+15\text{pF}$ (Probe) $\quad C_4 = 3.3\text{pF}$ $\quad C_5 = 22\text{pF}$

3 Experiment 2: Frequency Response

By the appropriate AC measurement techniques, fin the frequency response of the amplifier. Employ a frequency range of 10Hz to 100kHz. In addition complete the following tasks:

- Find the mid-band gain $\left|G_{v(\text{mid})}\right|$
- Find the bandwidth by measuring f_L and f_H and compute the Gain Bandwidth Product (GPB) which has units of Hertz, therefore make sure to convert the midband gain to V/V.
- Plot the measured results versus the simulated results on the same curve, make sure to use a magnitude range of 0 to +30dB
- Complete the summary tables at the end of the handout

DC Operating Point				
Device	Quantity	Simulated	Measured	Units
Q_1	I_D			μA
	$\lvert V_{OV} \rvert$			V
	V_G			V
	V_D			V
	V_S			V
Q_2	I_D			μA
	$\lvert V_{OV} \rvert$			V
	V_G			V
	V_D			V
	V_S			V
Q_3	I_D			μA
	$\lvert V_{OV} \rvert$			V
	V_G			V
	V_D			V
	V_S			V

Table 1: DC Summary Table

AC Summary			
Quantity	Simulated	Measured	Units
$\left\|G_{v(\mathrm{mid})}\right\|$			V/V
$\left\|G_{v(\mathrm{mid})}\right\|$			dB
f_L			Hz
f_H			kHz
BW			kHz
GBP			kHz

Table 2: AC Summary Table